Picking Floor Digital Twin: Data-Driven Warehouse Slotting

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Motivation/ Introduction

In a warehouse, an order picker spends more time **walking** than picking: in our model 79 s of every 150 s order is walking. Where each product sits on the shelves (**slotting**) decides that distance, yet it is usually set by habit. We built a **digital twin** of a picking floor that replays real customer orders, so any shelf arrangement can be measured before anyone moves a carton.

SCOPE of the Project

A 100-slot pick zone (5 aisles, racks on both sides) is driven by **2,293 real orders** from the UCI Online Retail II dataset (Nov 2010). We compare four slotting strategies, batch and smart-batch picking, and a **team of up to six pickers with aisle congestion**. The twin runs in a browser as a 3D game where users swap racks against an optimizer, and it emits the events real sensors would stream: RFID dock gate, UWB cart tags, pick-to-light modules and barcode scans.

Methodology

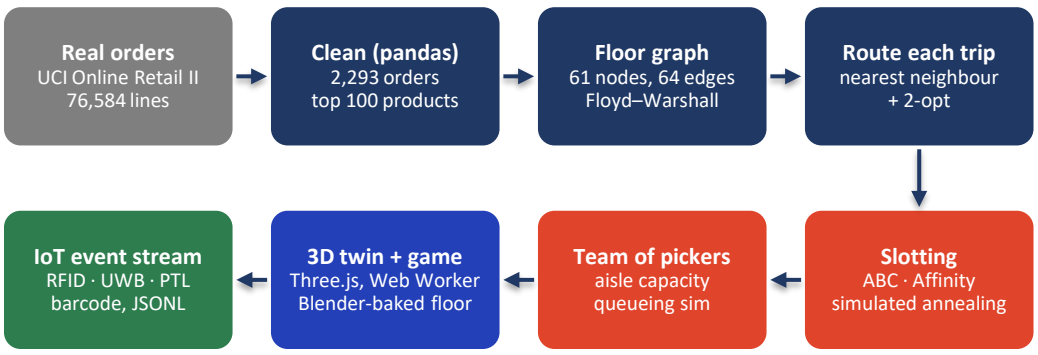


Fig. 1 System pipeline: one simulation core, used offline (Node) and live (browser).

a) Data. Cancelled invoices, postage/fee codes and invalid rows are removed (76,584 → 74,757 lines). The top 100 products by pick lines (3% of the range, 22% of picks) fill the 100 slots; 2,293 orders remain, 7.1 lines each.

b) Floor and routing. The floor is a graph whose edges follow aisles, so no route can cut through a rack. Floyd–Warshall gives all shortest paths; each trip (dock → picks → dock) is ordered by nearest neighbour and improved by 2-opt. Walking 1.2 m/s, 10 s per order line.

c) Slotting. Original (stock-code order); ABC (most-picked nearest the dock); Affinity (slot score = picks + w × co-orders with products already in that aisle); Optimized (simulated annealing over slot swaps, 80,000 iterations, accepting a worse swap with probability $P = e^{(-\Delta d / T)}$; only orders touching the two swapped products are re-routed).

d) Team of pickers. N pickers (1–6) clear the month's backlog from one queue. An aisle holds at most c carts (narrow: c = 1, wide: c = 2); a picker reaching a full aisle waits at its entrance. Time-stepped (0.5 s) and deterministic, so the browser and Node give identical results. Congestion loss = 1 − throughput / (N × single-picker throughput).

e) Validation. Layouts were also built from orders before 22 Nov only and tested on the 766 orders of the final week.

f) IoT layer. Every replay emits the events real devices would publish: RFID gate at the dock (cart out/in), UWB tag position at every aisle node, pick-to-light ON/OFF per slot, barcode pick confirmation and aisle-blocked alerts, exported as JSON Lines. An ESP32 publishing the same messages (e.g. over MQTT) could replace the replay feed.

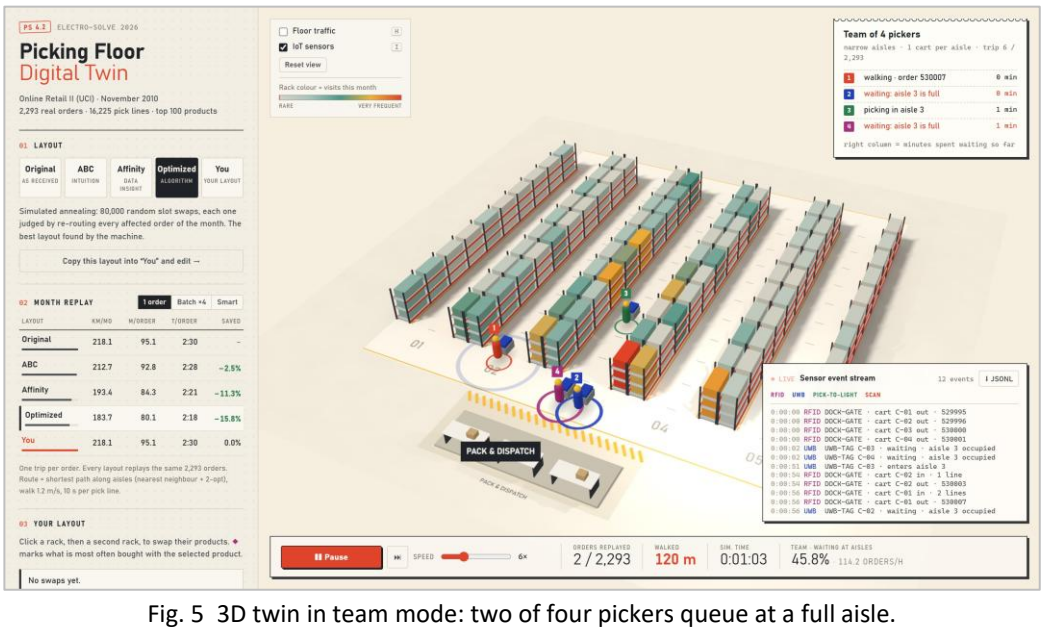


Fig. 5 3D twin in team mode: two of four pickers queue at a full aisle.

Results

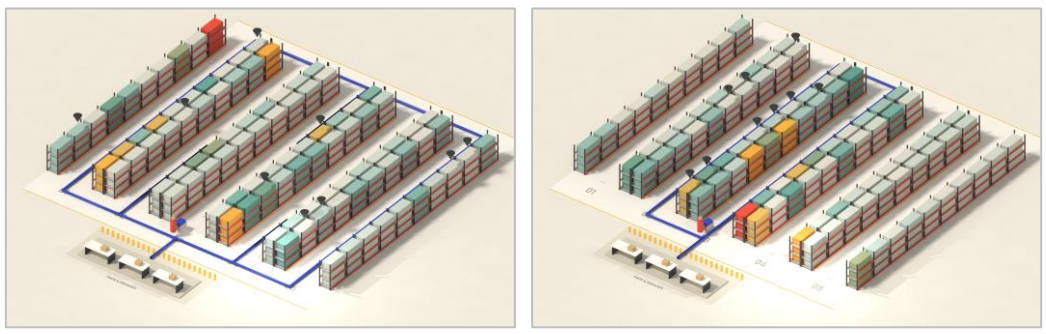
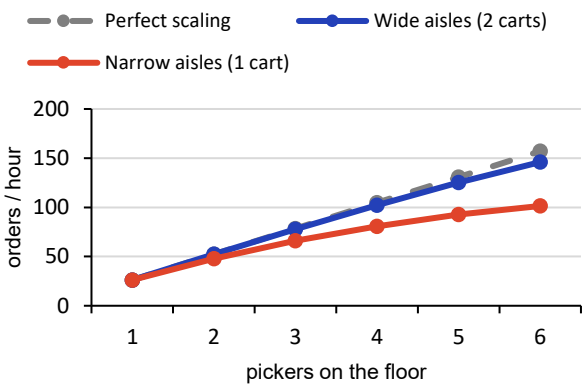


Fig. 2 One real order (invoice 532964, 9 lines): 143 m through 5 aisles on Original (left) vs 62 m through 2 on Optimized.

Layout	km / month	m / order	Walking saved	Unseen week
Original	218.1	95.1	–	–
ABC (intuition)	212.7	92.8	2.5%	1.9%
Affinity (data)	193.4	84.3	11.3%	11.2%
Optimized (SA)	**183.7**	**80.1**	**15.8%**	**13.9%**
Smart batch + Optimized	70.1	30.6	67.9%	–

Table 1 Same 2,293 orders on every layout, one picker. Unseen week: layouts built before 22 Nov, tested on the last 766 orders.



4 pickers, narrow aisles
23% of team capacity lost; pickers wait 23% of the time.

6 pickers
month cleared in 22.6 h (narrow) vs 15.7 h (wide aisles).

Slotting vs congestion
Optimized beats Original by +9% with 1 picker, +5% with 6.

Fig. 3 Throughput vs team size on the Optimized layout.

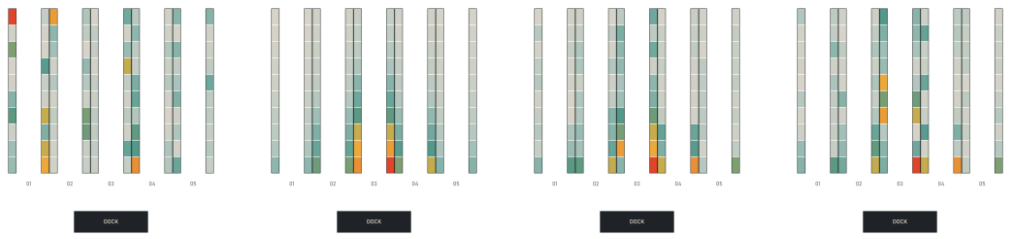


Fig. 4 Rack visits per month (grey = rare, red = very frequent); dock at the bottom.

SDGs Addressed:

- **SDG 8** Decent work: less walking, less fatigue
- **SDG 9** Industry & innovation: digital twin + IoT
- **SDG 12** Responsible production: leaner logistics

Conclusion/ Summary

The textbook ABC rule saves only 2.5% because orders average 7 lines spread across the floor. Using **what customers buy together** saves 11.3%, and simulated annealing 15.8% (34 km ≈ 8 picker-hours a month); the gains hold on unseen orders (13.9%). Smart batching plus the optimized layout cuts walking by 68%. With a **team**, narrow aisles lose 23–35% of capacity to queueing, which erodes about half the slotting gain: aisle width and batching matter as much as slotting. Next: live sensor input and dynamic re-slotting.

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Acknowledgments/ References

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